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PAPER

Ground motion modeling and epistemic uncertainty using conditional variational autoencoders

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Abstract

Reliable ground motion models (GMMs) are essential for accurate seismic hazard assessment. nonlinear principal component analysis (NLPCA) has been previously used for GMM development due to its capability in dimensionality reduction. This study evaluated the effectiveness of NLPCA in reducing response spectra dimensionality and predicting spectral ordinates using a large dataset from the engineering strong motion database. While NLPCA effectively transformed spectral ordinates into three principal components, instability in the computed components limited its predictive reliability, necessitating an alternative approach. To overcome these limitations, a conditional variational autoencoder (CVAE) integrated with an artificial neural network (ANN) was explored for GMM development. The CVAE + ANN framework enhanced predictive stability and enabled the generation of ensemble outputs for given set of predictor variables, effectively capturing epistemic uncertainty. The model incorporated key seismological parameters, including earthquake magnitude, source-to-site distance, shear wave velocity, focal mechanism, and focal depth. The proposed model successfully captured critical seismic characteristics such as magnitude scaling, ground motion attenuation, and nonlinear soil amplification, demonstrating predictive performance comparable to, and in some cases surpassing, existing GMMs. Additionally, the CVAE-based approach improved the representation of aleatory variability and epistemic uncertainty, offering a more robust and reliable framework for seismic hazard assessment.

1. Introduction

Advancements in strong motion instrumentation have increased the stock of ground motion records globally, aiding ground motion model (GMM) development for diverse seismic events. Both parametric and non-parametric GMMs have been developed in the past. Parametric models assume a predefined functional form between input and the target variable. These models often stem from physical principles, empirical data, or statistical assumptions. Parametric models excel when the problem's physics is well-defined due to their explicit functional dependencies. However, they struggle with complex, highly non-linear interactions. Akkar *et al* (2014a) used the RESORCE dataset to develop a parametric GMM for Europe and the Middle East, employing regression with mixed effects and bootstrap analysis to handle data uncertainty. Kotha *et al* (2020, 2022) utilized the engineering strong motion (ESM) dataset to create a parametric GMM for the European region taking into account the effects of magnitude (M_w) and Joyner–Boore distance (R_{JB}).

In contrast, ML-based models do not impose a fixed functional form. Instead, they learn patterns and dependencies from the data through iterative optimization and feedback mechanisms. These models capture complex non-linear relationships missed by parametric models. Certain ML-based approaches such has such as Bayesian neural networks (BNN), conditional variational autoencoders (CVAE), and conditional generative adversarial networks (CGAN) extend deterministic learning by modeling the probability distribution of the target variable rather than producing a single-valued prediction. ML-based models require constraints for robustness and interpretability. They rely on large datasets, lacks physical transparency, need uncertainty calibration, and are computationally intensive. However, their capacity to model non-linearity, process vast data, and provide probabilistic estimates surpasses parametric models, with ongoing advancements enhancing their reliability. Numerous ML-based GMMs have been proposed in several studies. Derras *et al* (2016) used the RESORCE database to develop artificial neural network (ANN)-based GMMs with Quasi-Newton backpropagation and regularization to prevent overfitting, refining parameters via maximum likelihood estimation. Recently, Sriwastav *et al* (2024) proposed a GMM for several ground motion parameters, as Yedulla *et al* (2025) proposed an ANN-based model for displacement response spectra and Fling. Mohammadi *et al* (2023) used ANN and extreme gradient boost (XGBoost) methods to predict peak ground acceleration (PGA), peak ground velocity (PGV), and psuedo spectral acceleration (PSA). Karimzadeh *et al* (2024) developed an ANN-based GMMs for PGA, PGV, and PSA using a set of recorded and stochastically simulated ground motions for the Turkish region. Ding *et al* (2025) developed a hybrid GMM to predict power spectral density across 0.1–25 Hz using 12 ML-based algorithms. They selected the top 7 models— support vector regression (SVR), RF, ERT, GBDT, XGBoost, LightGBM, and DNN for proposing the hybrid-based model. Similarly, Kuran *et al* (2024) used 6 different ML-based algorithms for predicting the PGV. Kang *et al* (2024) developed GMMs using the Kik-Net dataset to predict the PSA. Comparison between the parametric and non-parametric analysis was carried out, which concluded that the non-parametric model better captures the complex relationships present in the data. In general, ML-based methods demonstrated excellent capabilities in handling large datasets, and reducing error variance.

The correlation between periods of spectral ordinates poses a critical challenge for accurate assessment of seismic hazard, underscoring the need for alternative representations of uncorrelated variables. Dimensionality reduction techniques, such as principal component analysis (PCA) and its advanced variant, nonlinear PCA (NLPCA), have emerged as a powerful tool for addressing these complexities, enabling efficient and compact representation of high-dimensional ground motion data. NLPCA has found utility across various domains whenever there's a requirement for dimensionality reduction. NLPCA has been widely applied in structural health monitoring. Datteo *et al* (2017) proposed a long-term monitoring approach to model stadium stand vibrations using PCA to represent the structure's state in a low-dimensional space. It was observed that principal scores efficiently captured structural changes, with the first PC being especially sensitive to weather variations. The other components showed a more pronounced dependence on operational conditions measured through RMS signal excitation. Ye *et al* (2020) combined NLPCA with SVR in ambient effect filtering of high-rise buildings. The model frequencies of a structure are affected by correlated ambient factors such as temperature, wind speed, etc. The correlation among these factors is reduced through principal components (PCs) and were fed to an SVR model and trained on Benchmark building data with specific environmental conditions. The trained model was demonstrated to filter noise and eliminate the effect of ambient noise effectively. Ozdagli and Koutsoukos (2019) compared PCA and autoencoders (AE) for damage detection by using modal parameters, such as natural frequencies and mode shapes, to detect critical changes in structures under temperature uncertainty. Reconstructing these features, it was aimed to identify damage reliably. Through evaluation with structural response data, it was shown that integrating mode shapes significantly enhances damage detection, especially for minor damages.

In the context of GMM, Dhanya and Raghukanth (2022) utilized NLPCA for dimensionality reduction and response spectra prediction within GMM using the NGAWest2 database. It was shown that three PCs adequately represented the 91 spectral periods. An ANN model was also trained to predict these components, incorporating input parameters such as magnitude, distance, shear wave velocity, and focal mechanism. Esfahani *et al* (2021) investigated the dimensionality reduction of Fourier amplitude spectra (FAS) using AE using ground motions from the ESM database using M_w 3.5–7.0 and epicentral distances between 10 and 100 km. It was demonstrated that GMM based on a few PCs, considering magnitude, distance, and site class, can effectively capture the entire FAS. On a similar line, Basu *et al* (2024) conducted dimensionality reduction and feature extraction of ground motions using an ANN model based on NLPCA. It was shown that the intrinsic dimensionality of 100-dimensional FAS of acceleration time histories is three, with the first three PCs explaining at least 86% of the data.

Despite the distinct advantage of dimensionality reduction, the NLPCA algorithm and hence its predictions have been often observed to be unstable and non-unique (Lu and Pandolfo 2011). Bishop (1995) attributed the instability and inconsistency in NLPCA results to the optimization process used in training neural networks. Specifically, NLPCA relies on gradient descent-based techniques, which are highly sensitive to weight initialization. Since the NLPCA objective function is non-convex, gradient descent can lead to convergence at different local minima depending on the initial conditions, resulting in variations across multiple runs. Furthermore, the presence of complex, non-linear transformations in the network exacerbates this issue, making the model susceptible to overfitting or poor generalization. Kramer (1991) also highlighted key challenges in NLPCA, including non-unique solutions, where varying hyperparameters can lead to different yet equally valid reconstructions. Small input perturbations or architectural changes can significantly alter the extracted components, affecting reproducibility. Additionally, NLPCA is computationally intensive, requiring multiple training iterations and high complexity ($O(n^2k)$ or more) to solve non-linear optimization problems efficiently.

Variational AE (VAEs) (Kingma and Welling 2013) overcome NLPCA's limitations by adopting a probabilistic approach, ensuring a structured and unique latent representation through regularization. This mitigates non-uniqueness and enables synthetic data generation (ensemble prediction for a given set of seismological parameters) for augmentation and modeling. VAEs optimize reconstruction accuracy while maintaining computational efficiency. Mohammadi *et al* (2025) used VAE to generate synthetic data that fills gaps and augments underrepresented regions, improving shear resistance prediction for CFRP-strengthened RC beams. When incorporating seismological parameters like magnitude and fault mechanism, VAEs extend into CVAEs (Sohn *et al* 2015), integrating these variables into encoding and decoding processes. This enhances their ability to capture domain-specific variations and conditional relationships, making CVAEs a reliable alternative to NLPCA for high-dimensional seismic data modeling. Balmer *et al* (2024) used CVAE to solve forward and inverse design optimization problems by mapping inputs and design capacity onto latent space variables, then using them to derive optimized design parameters.

This study evaluates NLPCA for dimensionality reduction and spectral ordinate prediction using the ESM database and aims to propose a CVAE-based GMM. First, a discussion of ground motion dataset used is provided. Following this, the stability of NLPCA's PC is evaluated and the effectiveness of this approach (integrated with ANN) in GMM development is explored. Subsequently, a CVAE-based approach integrated with ANN is used to propose a GMM for the horizontal component with predictor variables as magnitude (M_w), Joyner–Boore distance (R_{JB}), site class (V_{s30}), focal depth and focal mechanism. Finally, the ensemble prediction using CVAE based approach is demonstrated and epistemic uncertainty is reported.

2. Ground motion database

In this study, ESM 2.0 database (Luzi *et al* 2020) is selected over the RESORCE dataset (Akkar *et al* 2014b) due to its broader coverage magnitude range and distances, which are critical for GMM development. The standardized processing of ground motion for ESM database (Puglia *et al* 2018) ensures high data reliability, making it more suitable for development of GMMs. While RESORCE is valuable for larger events (Derras *et al* 2016), ESM better aligns with the aim of this paper. The subset of ESM 2.0 used by Sriwastav *et al* (2024) is used in this paper. Moment magnitude (M_w) is sourced from the European Mediterranean Earthquake Catalog. Site classes are characterized by V_{s30} values derived from ESM2.0 flat file (Lanzano *et al* 2018). V_{s30} values in ESM2.0 are primarily based on *in-situ* geophysical measurements (obtained from https://esmdb.eu), with a considerable portion derived using the Multi-Channel Analysis of Surface Waves (MASW) method. In case *in-situ* measurements are unavailable, V_{s30} values are estimated from the slope, using the method proposed by Wald and Allen (2007). R_{JB} is used throughout this study to quantify source-to-site distance. The ground motion recordings lacking any of the key parameters such as M_w, R_{JB}, V_{s30}, focal depth (d), or fault mechanisms for any of the three components are excluded. A total of 11 675 3-component ground motions with $4.0 < M_w < 7.8$, $R_{\mathrm{JB}} < 400$ km, focal depth (d) < 40 km, V_{s30}: 180–1500 m s^{-1} are available after applying data exclusion criteria. The selected 11 675 ground motion recordings correspond to 1,464 seismic events and 1,402 recording stations distributed across Europe and its vicinity. Figure 1 presents the distribution of ground motion records with respect to M_w, R_{JB}, V_{s30}, focal depth ($\boldsymbol{D}$), and fault category. The data exhibits a non-uniform and skewed distribution, presenting challenges for linear regression analysis. Notably, there is also a scarcity of ground motion recording for the near-field region, and data within M_w 7.0–7.5 range, emphasizing the potential need for nonlinear regression techniques for improved predictions.

The correlation coefficient (Pearson correlation). $[\rho(\mathrm{Sa}(T_i), \mathrm{Sa}(T_j))]$, between target spectral ordinates (acceleration) across different periods are estimated for all 11 675 records. Figure 2(a) presents $[\rho(\mathrm{Sa}(T_i), \mathrm{Sa}(T_j))]$ where $\mathrm{Sa}(T_i)$ and $\mathrm{Sa}(T_j)$ are spectral ordinates at periods T_i and T_j. Distinct correlation is observed between the spectral ordinates at different periods with minimum correlation observed to be 0.4. A higher correlation between the spectral ordinates is observed when the logarithm of the spectral ordinates is considered as shown in figure 2(b). These plots highlight the redundancy and high dimensionality across 37 periods, motivating dimensionality reduction. NLPCA maps this high-dimensional, correlated space onto a lower-dimensional, uncorrelated space, preserving key spectral shape features for efficient representation, clustering, and sampling of ground motion spectra. Following section provides a brief discussion on NLPCA.

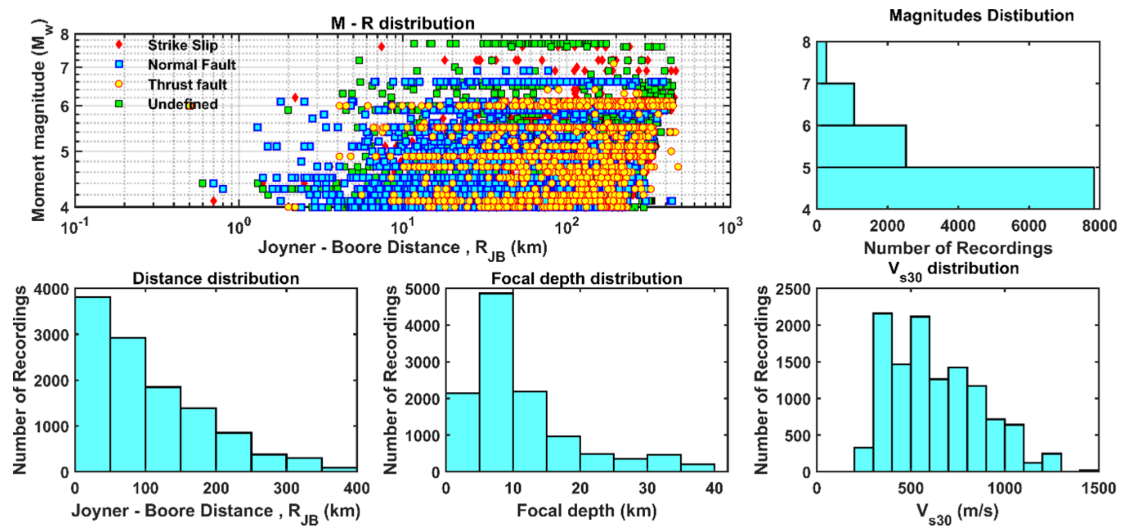


Figure 1. Distribution of recorded ground motion records with respect to seismological parameters.

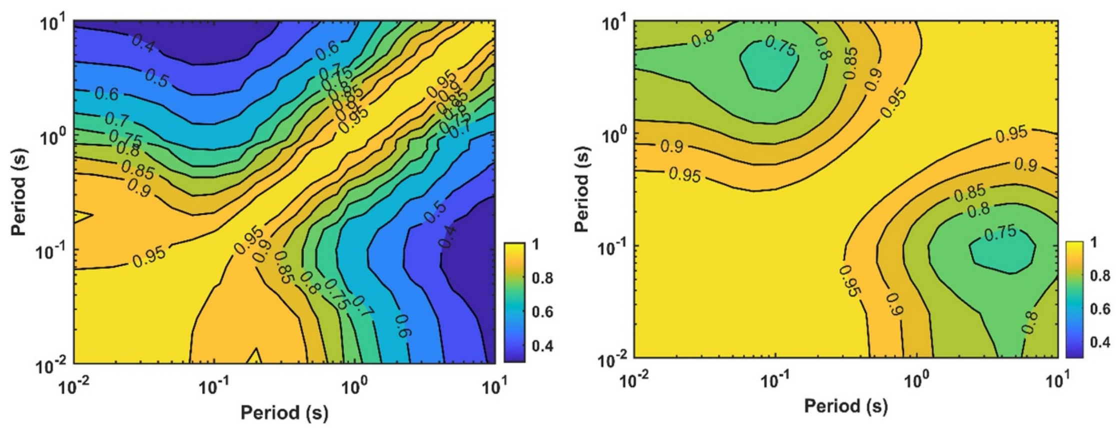


Figure 2. (a) The correlation coefficient between response spectra at different periods. (b) The correlation coefficient between the logarithm of response spectra at different periods.

3. NLPCA

Kramer (1991) pioneered the concept of NLPCA using principles of auto-associative neural networks. Unlike linear PCA, NLPCA represents PCs as curves, allowing accurate mapping of complex and non-linear data for dimensionality reduction. While a comprehensive methodology is provided by Hsieh (2009), a summary is presented here for clarity. A typical NLPCA network comprises of five segments. The first and the last segments represent the target spectral ordinates whose dimension needs to be reduced (figure 3). The third segment contains the dimensionally reduced PCs. The second and fourth segments contain one or more hidden layers. The second segment specifically reduces the dimensionality of the target spectral ordinate. The fourth layer can convert the PCs to a higher dimension. The first two segments project the input data to a reduced dimension bottleneck layer which is called as encoding layer. In figure 3, spectral ordinates at 37 periods taken as input variables (Y) are dimensionally reduced to 3 PCs. The form of PC_j can be expressed as

$$\mathrm{PC}_j = f_2\left[b_{j,2} + \sum_{k=1}^{m} W_{jk,2} f_1\left(b_{k,1} + \sum_{k=1}^{n} W_{ik,1} Y_i\right)\right] \quad (1)$$

Here, n and m represents the number of nodes in the input layer and first hidden layer, respectively. The parameters $W_{ik,1}$, $b_{k,1}$, are weights and biases between the input layer and the hidden layer. $W_{jk,2}$, and $b_{j,2}$ are weights, and biases between the first hidden layer and the bottleneck layer. f_1 and f_2 are the activation functions. The subsequent two layers map from the PCs back to the original dimension, aiming to

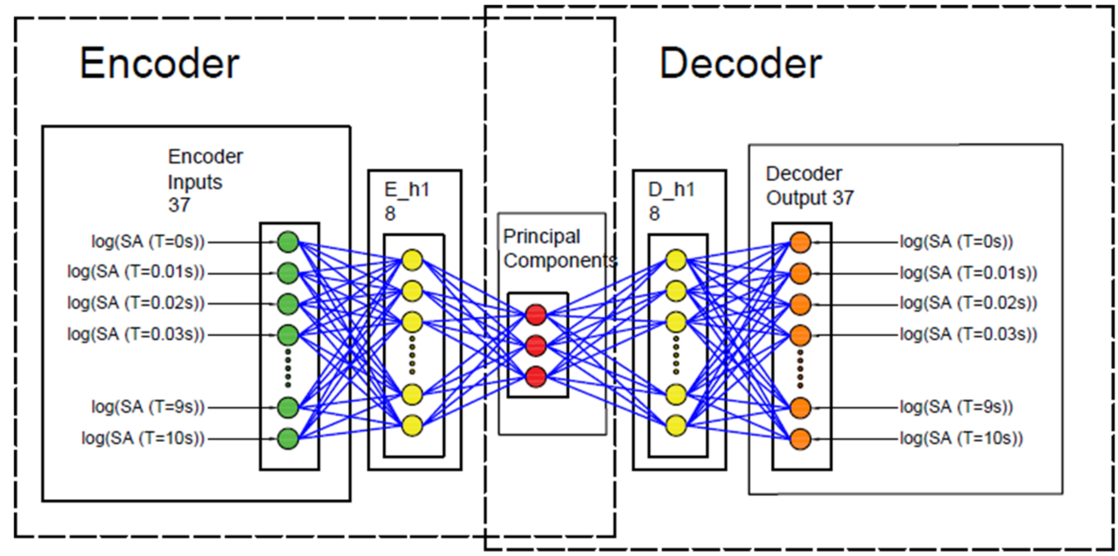


Figure 3. Nonlinear principal component analysis (NLPCA) architecture with 3 principal components.

reconstruct Y and is called the decoding layer. The detailed expression for the decoding layer can be written as

$$\hat{Y}_0 = f_4\left[b_{o,4} + \sum_{q=1}^{t} W_{oq,4} f_3\left(b_{q,3} + \sum_{j=1}^{s} W_{jq,3}\mathrm{PC}_j\right)\right]. \tag{2}$$

Here, $\hat{Y}_0$ represents the output data, s denotes the number of nodes in the bottleneck layer, and t represents the number of nodes in the final hidden layer. $W_{jq,3}$ and $b_{q,3}$ correspond to weights and biases between the bottleneck and the final hidden layer and that between the final hidden layer and the output layer is denoted by $W_{oq,4}$ and $b_{o,4}$. f_3 and f_4 are the activation layers. A combination of tanh and linear activation functions is observed to best represent the data based on trial and error and considered for f_1, f_2, f_3, and f_4. Other hyperparameters for the network are determined through mean square error (MSE) minimization between input and output values using the conjugate gradient descent algorithm.

The compressed section of auto-associative neural networks lacks consistent structure, with features randomly distributed across nodes. To address this, Scholz (2007) introduced hierarchical NLPCA (h-NLPCA), which ensures a variance hierarchy similar to linear PCA, where the first PC captures the highest variance. This can be achieved by constraining the variance in the component space or the L2 norm in the original space, effectively using error sub-networks to hierarchically enforce order. This approach generates uncorrelated components by eliminating complex correlations between them. In this study, h-NLPCA (referred to as NLPCA hereafter) is employed to derive the nonlinear PCs of logarithmic spectral ordinates at 37 periods ranging from 0 to 10 s. The 37 spectral periods align with widely used GMMs. To reduce the dimensionality from the 37 periods, NLPCA analysis is conducted. Initial trails are made to reduce the spectra to 2, 3, 5, and 10 PCs, selected through preliminary analysis to balance variance retention, model efficiency, and spectral feature preservation while minimizing redundancy. The sample network architecture for NLPCA (with 3 PCs) is illustrated in figure 3. A single hidden layer with the same number of hidden neurons is utilized in each of the 4 trial cases (2, 3, 5, and 10 PCs). In the figure 3, the number of neurons in the hidden layers responsible for mapping (m) and demapping (t) are kept identical. The NLPCA network in this study consists of five segments. The input layer (green) represents 37 spectral ordinates to be reduced, while the output layer (orange) reconstructs them after encoding and decoding. The latent space (red) representing PCs is aimed to capture most of the variance. Two intermediate layers, the encoder (E_h1) and decoder (D_h1), facilitate dimensionality reduction and reconstruction. The encoder compresses the 37-dimensional input into a latent space (Z), preserving key spectral variability, while the decoder reconstructs the spectral ordinates, ensuring an efficient representation of spectral shapes for further analysis. The number of neurons in hidden layers is crucial for capturing non-linearity. Too few may fail to model complexity, while too many risks overfitting.

$$\mathrm{SF} = \frac{0.1}{\max\left(\mathrm{std}\left(Y\right)\right)}. \tag{3}$$

Table 1. The percentage variance and cumulative percentage variance correspond to the NLPCA with different PCs.

	10PCs		5PCs		3PCs		2PCs	
Sl. no	Variance (%)	Cumulative variance (%)	Variance (%)	Cumulative variance (%)	Variance (%)	Cumulative variance (%)	Variance (%)	Cumulative variance (%)
1.	49.88	49.88	89.01	89.01	91.87	91.87	92.58	92.58
2.	21.80	71.68	9.04	98.05	7.36	99.23	7.42	100.00
3.	10.45	82.13	1.08	99.14	0.77	100.00		
4.	5.22	87.35	0.74	99.88				
5.	5.07	92.41	0.12	100.00				
6.	2.45	94.86						
7.	2.17	97.03						
8.	1.06	98.09						
9.	1.00	99.10						
10	0.90	100.00						

Table 2. Three principal components for different NLPCA trials demonstrating a non-unique solution.

Trial 1		Trial 2		Trial 3	
Variance	Cumulative variance	Variance	Cumulative variance	Variance	Cumulative variance
94.20	94.20	89.79	89.79	91.87	91.87
5.30	99.50	9.12	98.91	7.36	99.23
0.50	100.00	1.09	100.00	0.77	100.00

Table 3. Performance indices.

Index	Equation
Pearson correlation (P_{Cor})	$P_{\mathrm{Cor}} = \frac{\sum_{i=1}^{N} (y_i - \bar{y})\left(\hat{y}_i - \bar{\hat{y}}\right)}{\sqrt{\sum_{i=1}^{N} \left(\hat{y}_i - \bar{\hat{y}}\right)^2 \sum_{i=1}^{N} (y_i - \bar{y})^2}}$
Performance parameter (PP)	$\mathrm{PP} = 1 - \frac{\sum_{i=1}^{N} (y_i - \hat{y})^2}{\sum_{i=1}^{N} (y_i - \bar{y})^2}$
Error standard deviation $\sigma(\varepsilon)$	$\sigma(\varepsilon) = \sqrt{\frac{\sum_{i=1}^{N} \left(y_i - \hat{y}_i\right) - \overline{(\mathrm{y} - \hat{y})}}{N-1}}$

For computing variance of each trail networks (10, 5, 3, and 2 PCs), the spectral ordinates are first log-transformed and normalized to ensure comparability. The variance for each PCs is then quantified, and percentage variance is calculated by dividing each component's variance by the total dataset variance. Cumulative variance is determined by summing these percentages. While NLPCA is primarily used for dimensionality reduction, the variance metrics are derived from its non-linear framework. Table 1 presents the percentage of variance and the cumulative percentage variance captured by the NLPCA for different numbers of PCs (10, 5, 3, and 2 PCs). The results indicate that as the number of extracted PCs increases, the variance captured by the first PC decreases. This suggests that networks with fewer PCs are desirable as they result in simpler models while maintaining performance comparable to those with a higher number of PCs. Further, it is observed that the computed PCs vary in each of the NLPCA as shown in table 2. Three trials are shown for the network with 3PCs in table 2. The results indicate a non-unique solution of the NLPCA algorithm.

Despite non-unique, NLPCA provides a reliably good prediction. The modeling efficiencies of the four architectures are next assessed through parameters such as the Pearson correlation coefficient (P_{Cor}), performance parameter (PP) and error standard deviation, $\sigma(\varepsilon)$ (table 3). A model performs better when P_{Cor} and PP approach unity and $\sigma(\varepsilon)$ is low. The estimates for the four trial NLPCA network configurations analyzed in this study are summarized in table 4. It is observed that reducing the number of PCs led to comparatively higher estimates of R and PP and lower estimate of $\sigma(\varepsilon)$ upto 3PCs. The trend revered when only 2PCs are considered (table 4).

To evaluate the impact of dimensionality reduction on spectral reconstruction accuracy, the output responses of NLPCA networks with varying PC configurations is analyzed. Four separate NLPCA models are

Table 4. The performance of NLPCA with varying numbers of PCs.

Parameter	10PCs	5PCs	3PCs	2PCs
P_{Cor}	0.995	0.996	0.997	0.994
PP	0.991	0.993	0.993	0.986
$\sigma(\varepsilon)$	0.051	0.051	0.05	0.08

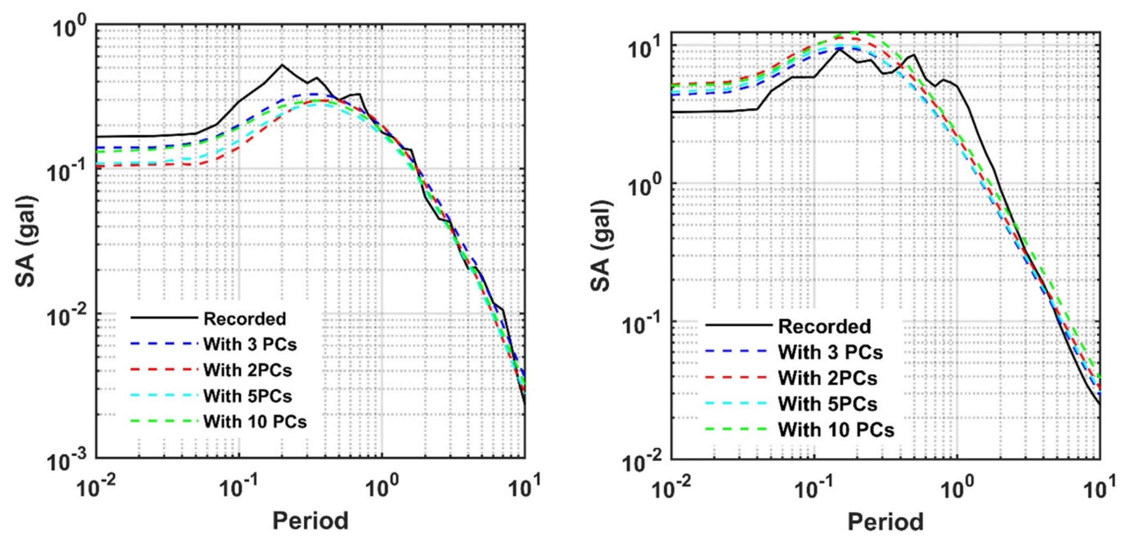


Figure 4. Comparison of the predicted response spectra using NLPCA with varying numbers of PCs against recorded spectra for two sample ground motions, (a) M_w 5.1, R_{JB}: 298.5, thrust fault, depth: 9.1 km and V_{s30}:804 m s^{-1}. (b) M_w 5.1, R_{JB}: 60.1 km, strike–slip, depth: 2.0 km and V_{s30}: 839.66 m s^{-1}.

Table 5. Correlation between 3 PCs.

	PC1	PC2	PC3
PC1	1	−0.0043	−0.0015
PC2	−0.0043	1	0.0036
PC3	−0.0015	0.0036	1

trained, maintaining a consistent network architecture while varying the number of PCs (2, 3, 5, and 10). After training, spectral reconstruction is carried out for each network, generating predicted spectral ordinates. The reconstructed spectra from each configuration are then compared in figure 4 using two representative ground motion records to balance between dimensionality reduction and spectral fidelity. The model predictions are observed to vary with the number of PCs used in the network. The predictions using 3PCs led to predictions closest to the spectra of the recorded ground motions. The overall discussions led to the impression that the best prediction is achieved through 3PCs and is adopted for the rest of the paper. The correlation between the 3 PCs scores is given in table 5 which indicates that these PCs are uncorrelated.

The Pearson correlation coefficient between the seismological parameters and the PCs are presented in table 6. The plots illustrating the trends of PCs with seismological parameters (M_w, R_{JB}, V_{s30}, and focal depth) are presented in figure 5. All three PCs exhibit correlation with magnitude, though PC_3 shows a weaker correlation. A similar trend is observed for R_{JB}. PC_1 and PC_2 have minimal correlation with V_{s30}, whereas PC_3 shows a stronger correlation. Additionally, PC_1 and PC_2 exhibit negligible correlation with focal depth, while PC_3 shows a slight correlation. Given the dependence of PCs on the seismological parameters such as M_w, R_{JB}, V_{s30}, an ANN model is next trained between these seismological parameters and the three uncorrelated PCs. Although the fault mechanism contributes modestly to the overall model, it is critical for characterizing attributes such as directivity effects, radiation patterns, and amplitude variations. Omitting these parameters could obscure essential relationships, particularly for specific earthquake scenarios. Hence, focal mechanism is also considered in as the predictor variable. The details of the ANN model are presented in the following section.

Table 6. Correlation of different parameters with the PCs.

Parameters	PC_1	PC_2	PC_3
Magnitude (M_w)	0.63	−0.55	0.13
Distance (R_{JB})	−0.46	0.77	0.12
Shear wave velocity (V_{s30})	0.09	0.06	0.44
Focal depth (d)	0.01	−0.02	−0.22

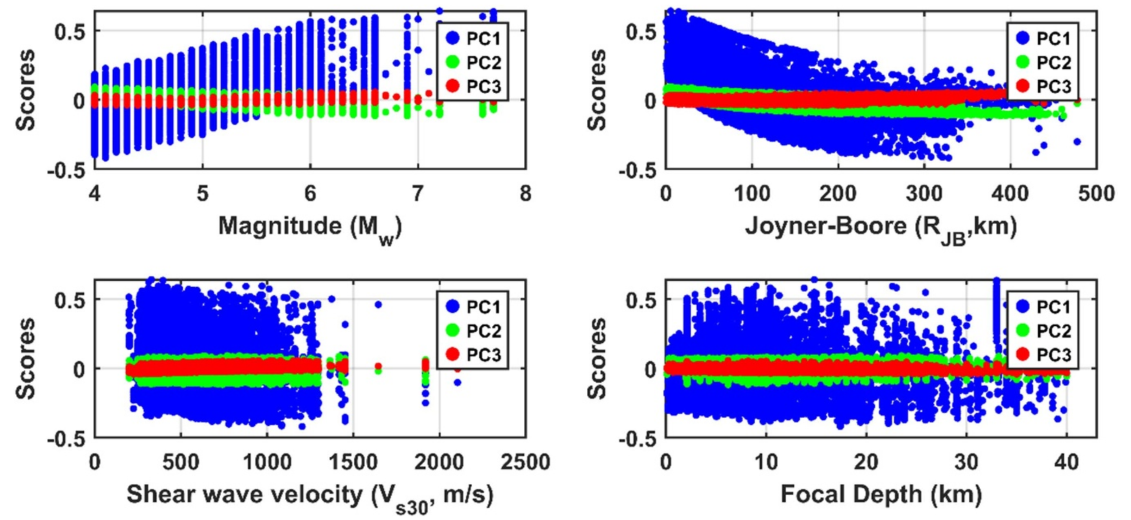


Figure 5. Dependence of the principal component scores in the non-linear-PCA analysis constituting three PCs with magnitude, distance, and shear wave velocity.

4. Development of ANN model for prediction of PCs

The analysis in the previous section demonstrated that the spectral acceleration across 37 periods could be effectively represented by just three PCs, which have a distinct correlation with the seismological characteristics. It is next aimed to predict these PCs as a function of seismological parameters such as M_w, R_{JB}, V_{s30}, fault mechanism (FM), and focal depth (d) as follows:

$$[PC_1,\ PC_2,\ PC_3] = f(M_w, R_{JB}, \log R_{JB},\ \log(V_{s30}),\ FM, d) + \text{error} \tag{4}$$

$PC_1,\ PC_2,\ PC_3$ represent the numerical values of the PCs for each ground motion record and are referred to as scores. These scores define the position of each record along the PCs axes. The architecture for predicting the principal components is depicted in figure 6. Numerous trials are made by varying the number of hidden layers, the number of neurons associated with each hidden layer, and the other hyperparameters such as random number generator to set initial weights, initial leaning rate, maximum number of epochs etc. First, the predictors as well as PCs, serving as target variables, are normalized to maintain consistency in network convergence. The dataset is divided into training (70%), validation (15%), and testing (15%) subsets and optimization of the hyperparameter is carried out. The optimized architecture features ELU activation layers, fully connected layers, and regression layers, utilizing a mini-batch size of 32 and a learning rate of 0.001 with random number generator set to 200. A single hidden layer with five hidden neurons is observed to satisfactorily obtain the desired physics characteristics of the ground motions and is adopted here. Using the predicted PCs, response spectra are reconstructed using the decoder of NLPCA. Using this ANN prediction followed by NLPCA reconstruction (NLPCA + ANN), a significant correlation (coefficient of determination. R^2) of 0.93 is noted between the target and predicted values. The NLPCA + ANN predictions are further evaluated and compared with the spectra of the recorded ground motions in the following section.

5. Performance of the model in the prediction of the spectral ordinate

This section evaluates the NLPCA + ANN model's capability to accurately represent the underlying physics in spectral ordinate prediction. The decoding layer is utilized to reconstruct response spectra across a range of seismological parameters using the predicted PCs. The variation of spectral ordinates with key seismic parameters—M_w, R_{JB}, focal depth and fault mechanism—is presented in the four panels of figure 7. Each

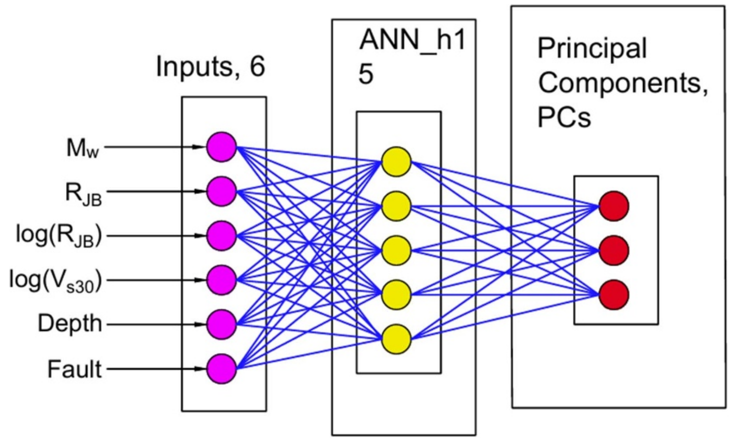


Figure 6. Developed ANN architecture to predict principal components of NLPCA with three PCs.

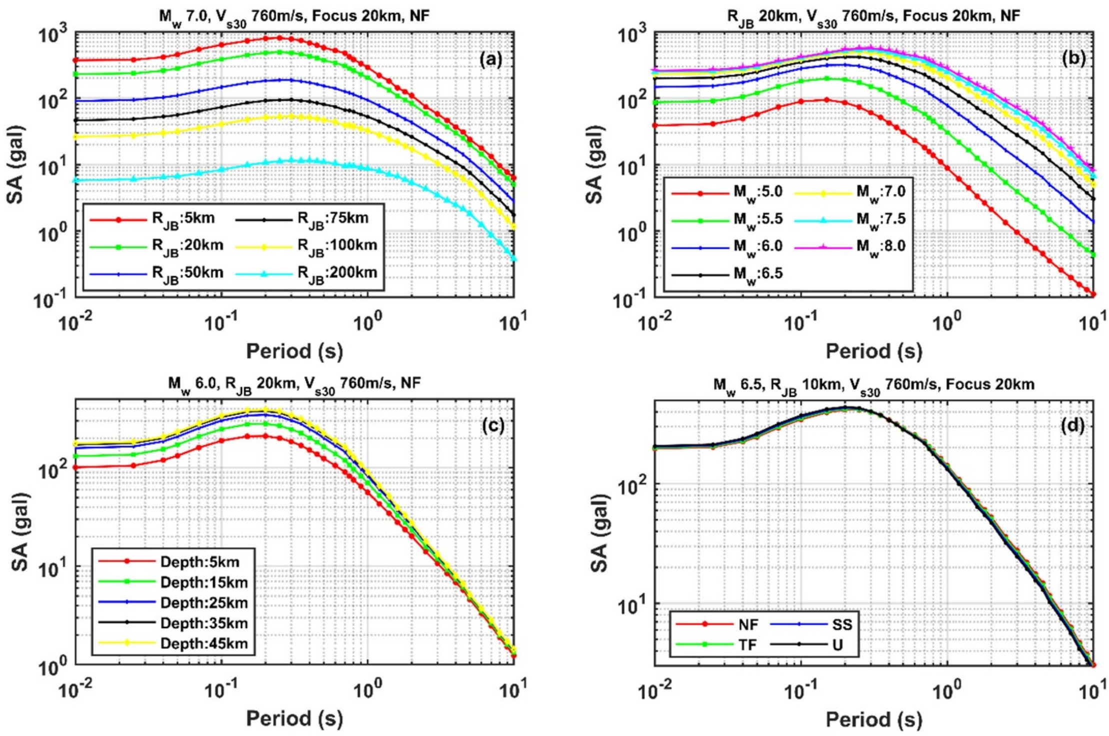


Figure 7. Variation of spectral acceleration with respect. (a) Joyner–Boore distance, b) magnitude, (c) focal depth and (d) fault mechanism (NF: Normal fault, SS: Strike slip, TF: Thrust fault and U: Unknown).

panel distinctly captures the shape of the response spectra and highlights the influence of these parameters on ground motion characteristics. In figure 7(a), the spectral acceleration decreases monotonically with increasing distance, effectively capturing the attenuation characteristics of ground motions. This trend follows seismological principles, where wave attenuation reduces spectral ordinates with distance. Figure 7(b) demonstrates the scaling of spectral ordinates with earthquake magnitude and the saturation effect at higher magnitudes. Higher magnitudes increase spectral accelerations, but the rate slows at larger magnitudes due to saturation. The influence of focal depth on spectral ordinates is presented in figure 7(c). Deeper focal depths correspond to higher spectral ordinates due to greater energy release and reduced attenuation in consolidated materials. In contrast, the impact of the fault mechanism on spectral ordinates, as shown in figure 7(d), is relatively small. NLPCA's instability may have limited its ability to capture fault mechanism effects, despite its effectiveness in dimensionality reduction.

To further explore the influence of site conditions, figure 8 presents the variation of spectral ordinates with V_{s30} for different magnitudes (M_w 5.5, 6.5, 7.0, and 7.5). The analysis is conducted under fixed conditions of R_{JB} 20 km, focal depth 20 km, and normal faulting style. As anticipated, lower V_{s30} correspond

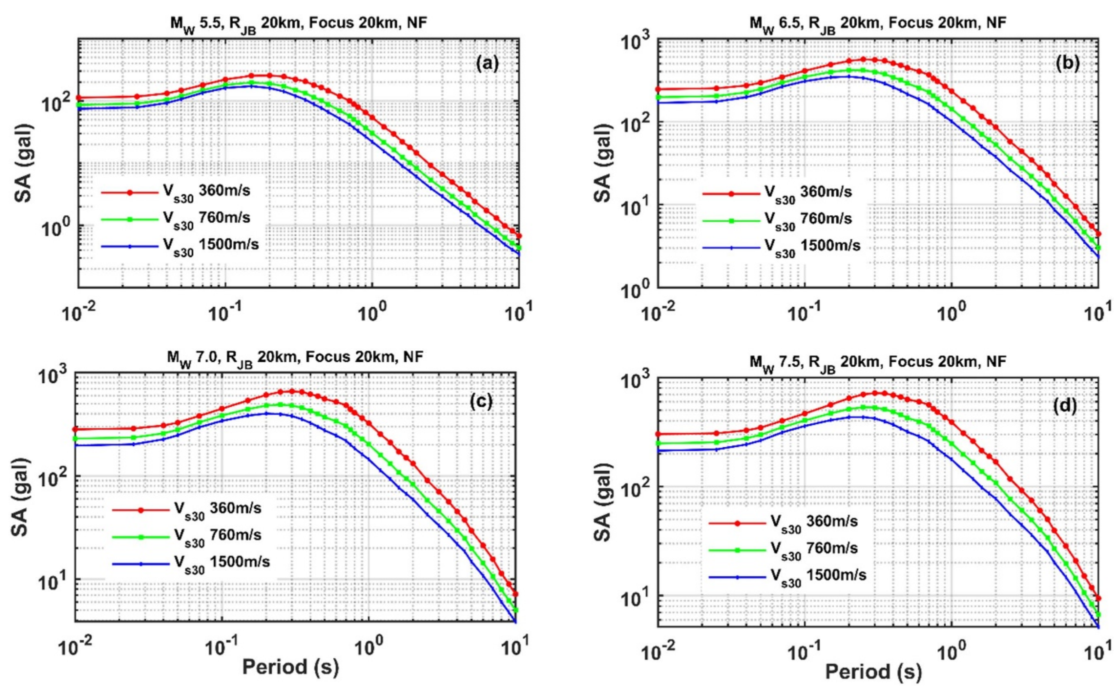
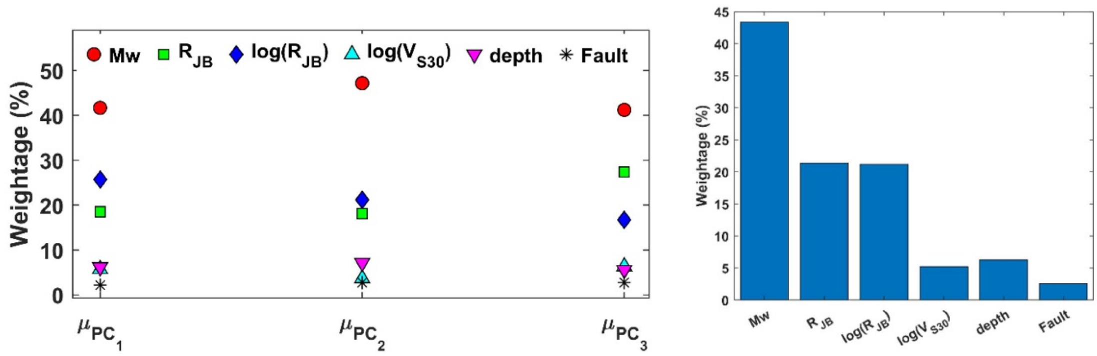


Figure 8. Variation of spectral acceleration, for M_w 6.0, 6.5, 7.0, and 7.5, distance and focal depth of 20 km, normal fault mechanism, and V_{s30} 760 m s^{-1}.



a) Contribution of predictors to each PCs

b) Contribution of predictors to the model

Figure 9. The relative impact of the input parameters in the NLPCA+ ANN model.

to higher spectral ordinates across all magnitudes. This finding underscores the amplification effect of softer soils, which tend to amplify seismic waves and result in stronger ground shaking compared to stiffer sites. The overall findings indicate that the methodology relying on just three uncorrelated PCs captures the desired ground motion characteristics. Figure 9 presents effect of predictors on each of the principal components (PCs). It can be observed that the M_w has maximum influence on each of the PCs followed by R_{JB}. In contrast, the fault mechanism has minimal influence.

5.1. Evaluation of the accuracy of the NLPCA predictions

The accuracy of NPLCA predictions following the ANN model for PCs is further evaluated in this section. The mean values of the performance metrics for NLPCA + ANN network across all the period indicate satisfactory model performance, with a low root mean squared error (RMSE) (0.162), mean absolute error (MAE) (0.198), and a high R^2 (0.93), reflecting strong predictive accuracy. However, the average mean absolute percentage error (MAPE) of 38% indicate greater relative errors in certain predictions leading to challenges in capturing extreme values. The comparison of response spectra for recorded ground motions with the reconstructed response spectra is presented for four sample ground motions in figure 10. The predicted (reconstructed) response spectra align well with the recorded spectra for the top two panels while there is a considerable difference in the alignment of reconstructed response spectra with recorded spectra in

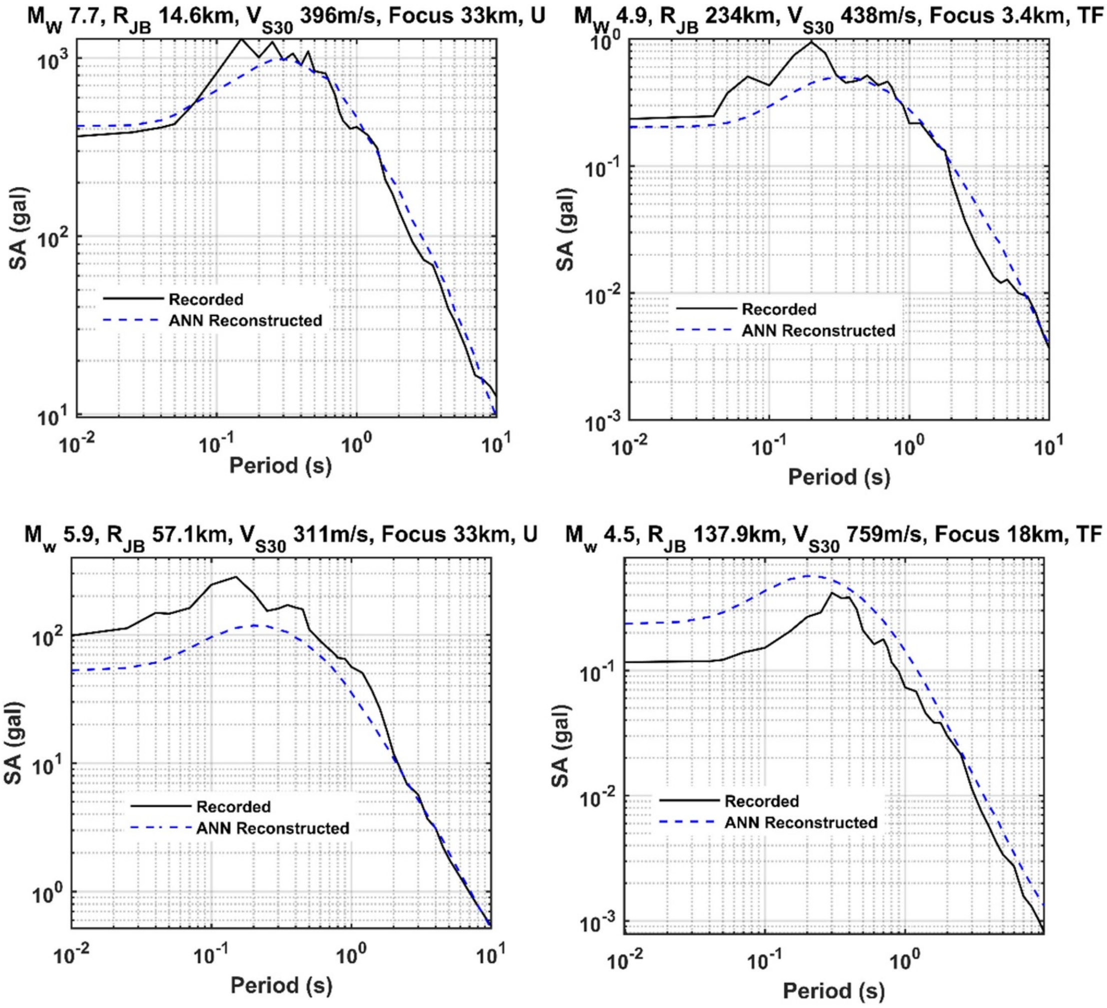


Figure 10. Comparison of the response spectra for recorded ground motions with the reconstructed response spectra using ANN model considering 3PCs for 4 sample ground motions.

the bottom two panels. Similar inconsistencies between the NLPCA prediction and the spectra of the recorded ground motion have been observed for numerous ground motions. This indicates that although performance parameters are satisfactory and as well as the desired seismological characteristics being satisfied, predictions may be quite far from the recordings in NLPCA. Further, as demonstrated earlier, NLPCA predictions lack stability, indicating the need for improved GMMs using dimensional reduction. In this regard, CVAE is explored to GMM development in the following section.

6. CVAE

A variational AE (VAE) generalizes the architecture of conventional AE and NLPCA by incorporating a probabilistic formulation for latent space modeling. In contrast to deterministic AE, which map input data to fixed latent embeddings, VAEs introduce a prior probability distribution—commonly a multivariate Gaussian with zero mean and unit variance—over the latent variables. This probabilistic structure is enforced through a variational inference framework, where the encoder network approximates the intractable true posterior distribution of the latent variables given the input data, while the decoder network reconstructs the data from samples drawn from this latent distribution.

The optimization objective of a VAE combines a reconstruction loss, ensuring fidelity of the decoded outputs to the input data, with a KL divergence term, which regularizes the learned latent distribution to align with the prior. This dual objective enables VAEs to not only capture the underlying data manifold but also to generate novel data points through controlled sampling and interpolation in the latent space. The VAE models two distributions: the variational approximation distribution $q(z \mid x;\theta)$ and the generative model $p(x \mid z;\rho)$, where x denotes the input and z the latent variable. In this setup, $q(z \mid x;\theta)$ serves as the encoder, transforming observable variables into latent variables, while $p(x \mid z;\rho)$ serves as the decoder,

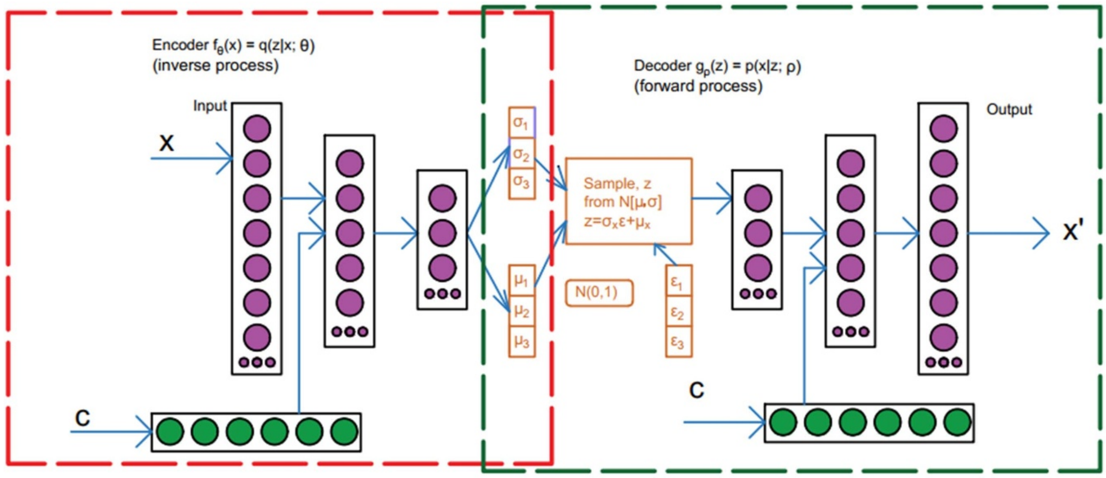


Figure 11. Schematic representation of the conditional variational autoencoder (CVAE) architecture, where X denotes the spectral ordinates used for training and C represents the conditional inputs.

converting latent variables back into observable variables. The CVAE extend VAEs by conditioning the latent space and reconstruction on additional variables (e.g. seismological parameters). This conditioning is integrated into both the encoder and decoder networks, allowing CVAEs to model conditional distributions and capture domain-specific variations on ground motion predictions. The typical architecture of a CVAE is depicted in figure 11.

CVAE works on the same principle as NLPCA with the additional advantage of computing the probabilistic estimate of PCs rather than a deterministic estimate. In case of the CVAE encoder, the hidden layer representation h is derived from the encoding function. The deep encoding function with n hidden layers can be written as

$$h = f_n\left(\left(\ldots f_2\left(W^{(2)} f_1\left(W^{(1)}x\right) + b^{(1)}\right) + b^{(2)} + \ldots\right) + b^{(n)}\right) \tag{5}$$

where W and $\boldsymbol{b}$ represents weight and bias, respectively, with the superscript (k) represents 'k^{th} 'layer. The function $f_k(x)$ is the activation function for the k^{th} hidden layer. The goal of training the model is to ensure that the encoder generates hidden variable vectors that follow a Gaussian distribution. The KL divergence is used to quantify the difference between the normal distribution $N\left(\mu,\sigma^2\right)$ of the hidden variables and the Gaussian distribution $N(y,1)$. The KL loss function of the encoder is:

$$J_1\left(\boldsymbol{\mu},\boldsymbol{\sigma}^2;\boldsymbol{y}\right) = \mathrm{KL}\left(N\left(\boldsymbol{\mu},\mathrm{diag}\left(\boldsymbol{\sigma}^2\right)\right)||N(\boldsymbol{y},\boldsymbol{I})\right). \tag{6}$$

Bold symbols represent vectors. $\mathrm{KL}\left(N(\boldsymbol{\mu},\mathrm{diag}(\boldsymbol{\sigma}^2)) \parallel N(\boldsymbol{y},\boldsymbol{I})\right)$ is the KL divergence from $N(\boldsymbol{\mu},\mathrm{diag}(\boldsymbol{\sigma}^2))$ to $N(\boldsymbol{y},\boldsymbol{I})$. $N(\boldsymbol{\mu},\mathrm{diag}(\boldsymbol{\sigma}^2))$ represents the multivariate Gaussian distribution with a mean vector $\boldsymbol{\mu}$ $(\in \mathbb{R}^m)$ and diagonal covariance matrix $\mathrm{diag}(\boldsymbol{\sigma}^2)$ (where $\boldsymbol{\sigma}^2$ $(\in \mathbb{R}^m)$) of the approximate posterior distribution indicating independent latent variables. m is the dimensionality of the latent space (i.e. the number of latent variables). $N(\boldsymbol{y},\boldsymbol{I})$ represents the standard multivariate normal distribution used as the prior, with mean vector $\boldsymbol{y}$ $(\in \mathbb{R}^m)$ (typically set to 0) and identity covariance matrix $\boldsymbol{I}$. Expanding the KL divergence for a multivariate Gaussian with a diagonal covariance matrix, we get:

$$J_1\left(\boldsymbol{\mu},\boldsymbol{\sigma}^2;\boldsymbol{y}\right) = \frac{1}{2}\sum_{i=1}^{m}\left(-\log{\sigma_i}^2 + \left(\mu_i - y_i\right)^2 + {\sigma_i}^2 + 1\right) \tag{7}$$

μ_i and ${\sigma_i}^2$ represents the mean and variance of i^{th} latent variable. It is necessary that the mean output from the encoder for a sample x_i be close to the sample y_i. The loss function of the encoder is as follows:

$$J_2\left(y_i,\hat{y}_i\right) = \frac{1}{m}\sum_{i=1}^{m}\left\|y_i - \hat{y}_i\right\|^2 \tag{8}$$

where m represents dimension of latent space. To generate the input for the CVAE decoder, a sample z need to be generated from $N\left(\mu_x,\sigma_x^2\right)$ using the re-parameterization trick. This involves sampling ε from $N(0,1)$ and then computing $z_x = \mu_x + \varepsilon\sigma_x$. The input space of the decoder is $z = \{z_1,\ z_2,\ldots,\ z_m\}$. The h' represents the hidden layer and is obtained by applying the decoding function of the decoder input. The decoding function can be expressed as

$$h' = g_m\left(\dots g_2\left(W'^{(2)} g_1\left(W'^{(1)} x\right) + b'^{(1)}\right) + b'^{(2)} + \dots\right) + b'^{(m)}\right). \tag{9}$$

The above equation is analogous to that for the encoder, with parameters $W'^{(1)}, b'^{(1)}, g_1(x), W'^{(k)}, b'^{(k)}, g_k(x)$ holding the similar meaning as in equation (5), but applied to the decoder. The loss function of the decoder, known as the reconstruction error, measures the discrepancy between the network input $x = \{x_1, x_2, \dots, x_n\}$ and the output. It is expressed as

$$J_3(x_i, \hat{x}_i) = \frac{1}{n}\sum_{i=1}^{n} ||x_i - \hat{x}_i||^2 \tag{10}$$

where n stands for the dimension of input space.

6.1. CVAE objective function

The re-parameterization is carried out for the encoder output (μ_x, σ_x), to obtain $z_i = \mu_x + \varepsilon\sigma_x$ (where $\varepsilon \in N(0,\ 1)$) as the input to the decoder. The framework of CVAE is formed by connecting the encoder and decoder. The network consists of 8 layers (figure 12). The first 4 layers account for encoders, which compress the input vector x (representing the spectral ordinate of the horizontal component between 0 to 10 s) to vector $\left(\boldsymbol{\mu}, \mathrm{diag}(\boldsymbol{\sigma}^2)\right)$ using its mapping function $\left(\boldsymbol{\mu}, \mathrm{diag}(\sigma^2)\right) = f_\theta(x)$, where θ is a parameter. Layers 5–8 are decoders, which decodes $z_x = \mu_x + \varepsilon\sigma_x$ into $\hat{x}$, and its mapping function is expressed as $\hat{x} = g_\rho(z)$, where ρ denotes the learnable parameters (such as weights and biases) of the decoder network. The vector has the same dimensions as the input x, and the network attempts to reconstruct x, that is, $x = \hat{x}$. Assuming $y = \hat{y}$, $\hat{y} = f_\theta(x)$ is regarded as an inverse function, and $\hat{x} = g_\rho(z)$ is a forward function. Based on this, it is necessary to make $x = \hat{x}$ and $y = \hat{y}$ and $N(\boldsymbol{\mu}, diag(\boldsymbol{\sigma}^2))$ to $N(\boldsymbol{y}, \boldsymbol{I})$ at the same time for network optimization. To this end, suppose the training set has n sample data $\{x_i, y_i\}$, where $i = 1,2, \dots, n$, then the objective function of the whole network can be expressed as:

$$\begin{aligned} J(\theta, \rho; x) &= \lambda_1 \times J_1\left(\boldsymbol{\mu}, \boldsymbol{\sigma}^2; \boldsymbol{y}\right) + \lambda_2 \times J_2(y_i, \hat{y}_i) + \lambda_3 \times J_3(x_i, \hat{x}_i) \\ &= \lambda_1 \times KL\left(N\left(\mu, \sigma^2\right) || N(\mu, 1)\right) + \lambda_2 \times \frac{1}{m}\sum_{i=1}^{m} ||(y_i - \hat{y}_i)||^2 + \lambda_3 \times \frac{1}{n}\sum_{i=1}^{n} ||(x_i - \hat{x}_i)||^2 \end{aligned} \tag{11}$$

λ_1, λ_2 and λ_3 are the weighting coefficients of the loss function in the encoding and decoding processes, respectively, and the gradient function is used to optimize the objective function $J(\theta,\ \rho;\ x)$. The weighting coefficients are empirically optimized through trials to balance KL divergence, latent space regression loss, and reconstruction error. A lower λ_1 (0.1) is provided to prevent excessive regularization, preserving meaningful latent representations. λ_2 (1.0) ensured latent variables capture relevant structures, while a higher λ_3 (10) is provided to prioritize the reconstruction accuracy.

6.2. CVAE architecture

In the case of CVAE, 99 target spectral ordinates associated with each period is considered rather than the 37 period to demonstrate its effectiveness in dimensionality reduction. The input is conditioned to six conditional features—M_w, R_{JB}, $\log(R_{\mathrm{JB}})$, $log(V_{s30})$, FM, and d. Network hyperparameters in CVAE include the number of hidden layers, the number of neurons per layer, activation functions, batch size, initial learning rate, and the number of epochs. Initial trials with various network configurations are conducted to finalize the number of neurons and hidden layers, aiming to minimize the objective function and maximize the correlation between the target and the prediction. Through numerous trials, the number of hidden layers in both the encoder and decoder is finalized as 2. The two fully connected hidden layers of the encoder network have 256 and 128 neurons, respectively for extracting latent representations, followed by layers for the mean and standard deviation of the latent space to enable reparameterization. The decoder, also with two hidden layers (128 and 256 neurons), receives input from both the latent representation and the conditional features, generating 99-dimensional regression outputs. Both encoder and decoder architectures are transformed into deep learning networks (*dlnetwork in MATLAB*) for efficient backpropagation and gradient updates.

The optimized network has a batch size of 32 and spans 300 epochs. The model's initial learning rate is 0.001, balancing steady convergence with stable parameter updates. The Adam optimizer is used to optimize parameter updates, and an exponential moving average is applied to stabilize the gradients. ELU activation functions are employed in both the encoder and decoder networks, specifically after each of the hidden

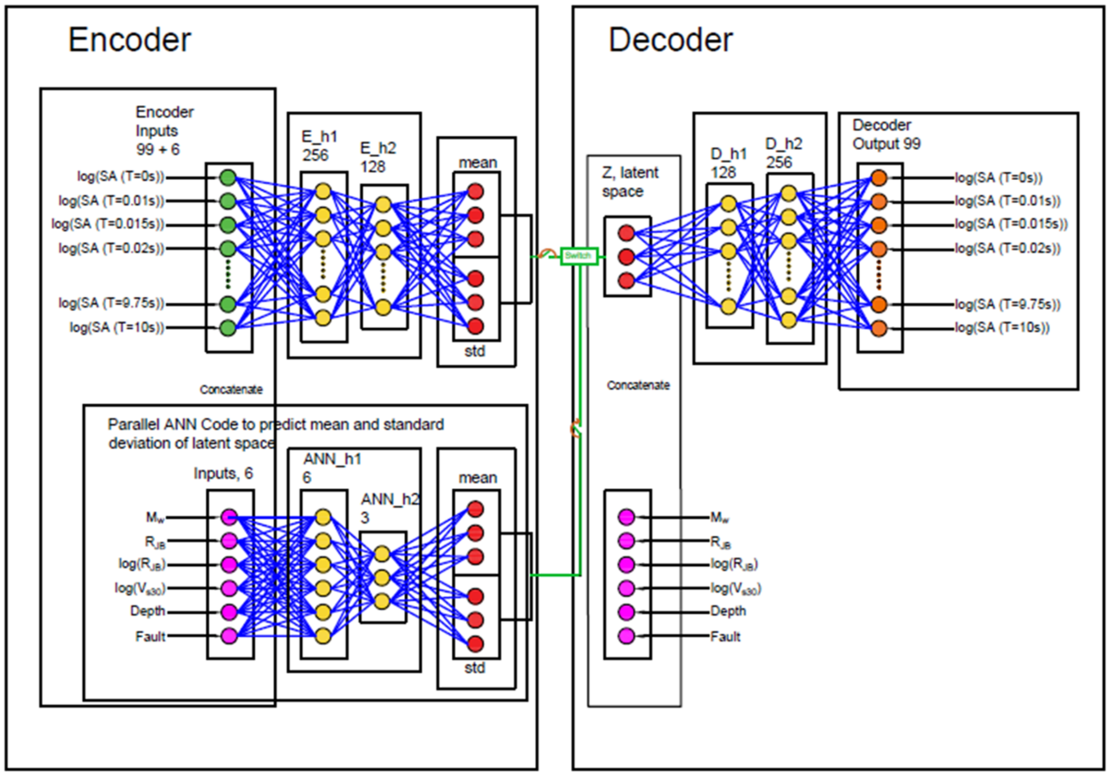


Figure 12. CVAE + ANN architecture with three 3 principal components.

layers. ELU functions introduce non-linearity into the network, allowing the model to learn complex representations by transforming negative inputs to zero and passing positive inputs unchanged. This activation helps prevent vanishing gradients, enabling efficient learning in deep networks. Weights are initialized with small random Gaussian values, while biases are set to zero. The reparametrize function injects randomness into the latent space, ensuring smooth gradient flow, stabilizing training, and enhancing model convergence. The objective function minimizes reconstruction error and regularizes the latent space for accurate predictions and stable dimensionality reduction. Final predictions show strong correlation with actual response spectra, aligning with seismic wave propagation physics and highlighting CVAE's effectiveness in capturing complex seismic responses. Since our aim to develop a GMM, CVAE is integrate with ANN. Spectral ordinates are encoded into PCs and trained against seismological parameters using ANN. The trained PCs are then decoded to reconstruct spectral ordinates, enabling ensemble generation for new seismic inputs. ANN model for mean and standard deviation of PCs is discussed in the following section.

6.3. ANN for training the mean and standard deviation of PCs

The mean and variance of the PCs obtained using the CVAE are further trained using ANN. Here the input parameters M_w, R_{JB}, V_{s30}, focal depth, and fault mechanism are normalized for effective training. The target variables, representing latent means and standard deviations, are similarly normalized to ensure consistency in network convergence. The data is split into training (70%), validation (15%), and testing (15%) sets. The optimized architecture incorporates a series of ELU activation layers, fully connected layers, and regression layers, configured with a mini-batch size of 32 and a learning rate of 0.001 with a random number generator of 200 for initializing the weights and biases. Two hidden layers with 6 and 3 neurons respectively provided the minimum MSE. The training utilized the ADAM optimizer with early stopping based on validation loss to mitigate overfitting. The network architecture of CVAE + ANN network is presented in figure 12.

7. Performance of the CVAE + ANN model

Post-training the CVAE + ANN model, the latent reparameterization step is used to generate samples from the mean and standard deviation vectors, reconstructing the response spectra through the decoder. The predictions are validated by comparing model output to actual spectral ordinates across various ranges of seismological parameters in figure 13. The CVAE decoder can generate a random ensemble of response spectra from the learned latent distribution. Such predictions, referred to as synthetic data, are generated

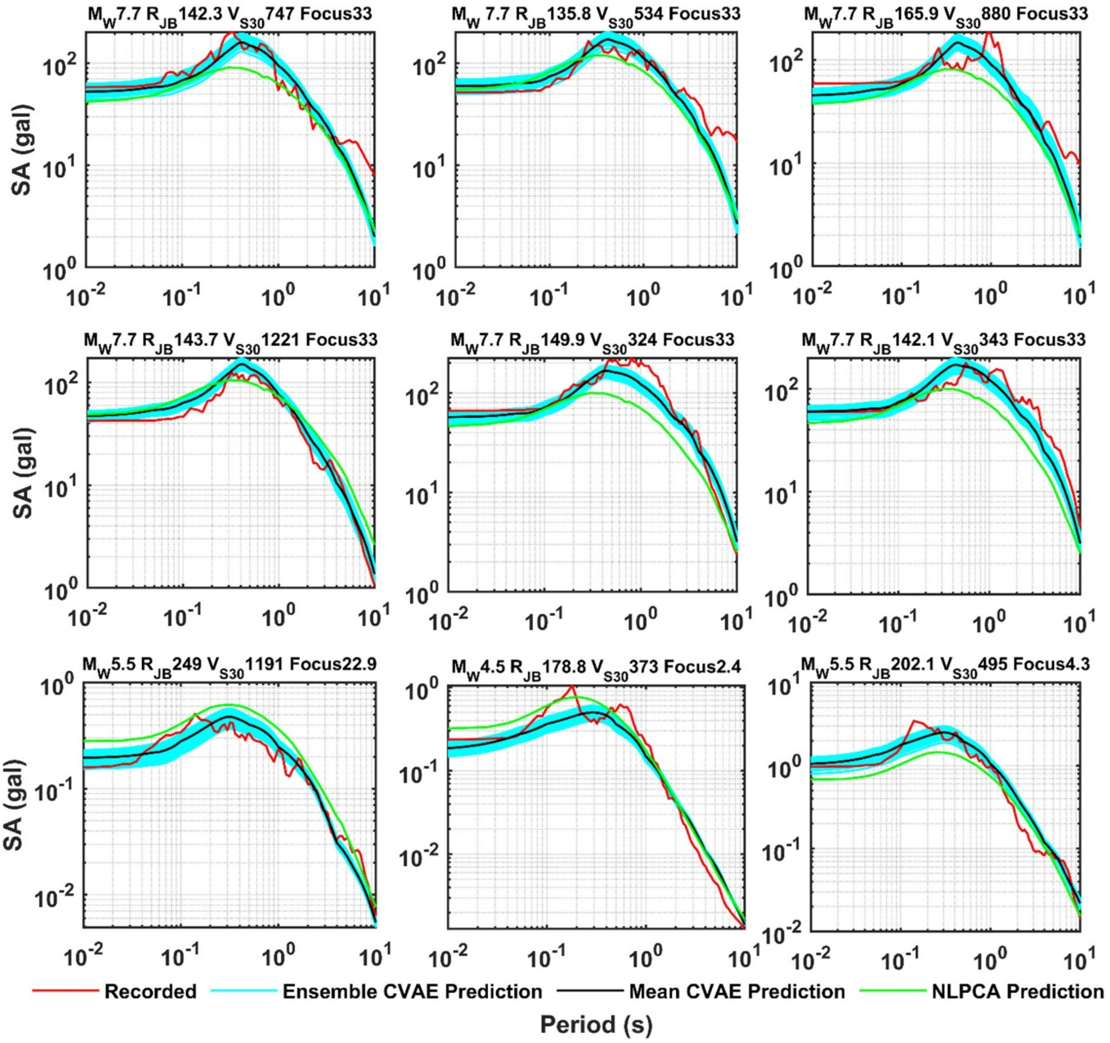


Figure 13. Comparison of the 100 ensemble CVAE + ANN prediction with the response spectra of the recorded events for 9 sample records and NLPCA prediction.

conditionally on specific input parameters (e.g. seismological conditions), enabling the creation of realistic response spectra. 100 Samples (represented by cyan lines) are generated for each ground motion and its median (black line) is compared to the response spectra of the recorded ground motion (red line) (figure 13). The comparison is presented for 9 randomly selected ground motions across different seismological parameters. Visualizations of the predicted spectral ordinates reveal the model's accuracy and potential for seismic response prediction, confirming the CVAE's ability to capture complex dependencies in earthquake data. The figure also includes the prediction of the NLPCA + ANN model discussed in sections 4 and 5 of the paper. Better predictive power of CVAE + ANN network is evident.

An initial assessment of the model performance is done using the correlation R^2 values between the recorded and predicted spectral ordinate. The performance metrics of the CVAE + ANN model demonstrate stronger predictive capability, with a low RMSE (0.136) and MAE (0.183) averaged across the periods, along with a high R^2 (0.95), indicating reliable accuracy in representing data variability. Furthermore, the MAPE (27%), averaged across the period, is comparatively lower compared to the NLPCA-based model, demonstrating CVAE's superior accuracy, enhancing spectral modeling reliability and robustness. Next, predictions are evaluated according to the fundamentals of earthquake physics. Figure 14(a) presents predicted response spectra for different R_{JB} for specific seismological parameters defined by M_w 7.0, V_{s30}: 760 m s^{-1}, focal depth 20 km, and Normal faulting. The expected attenuation property of the response spectra with distance is observed. Figure 14(b) presents sample response spectra predictions across different magnitudes with $R_{\mathrm{JB}} = 20$ m, $V_{s30} = 760$ m s^{-1}, and a focal depth of 20 km with normal faulting. The distinct increase in spectral amplitude with M_w confirms the magnitude scaling effect of the model. The

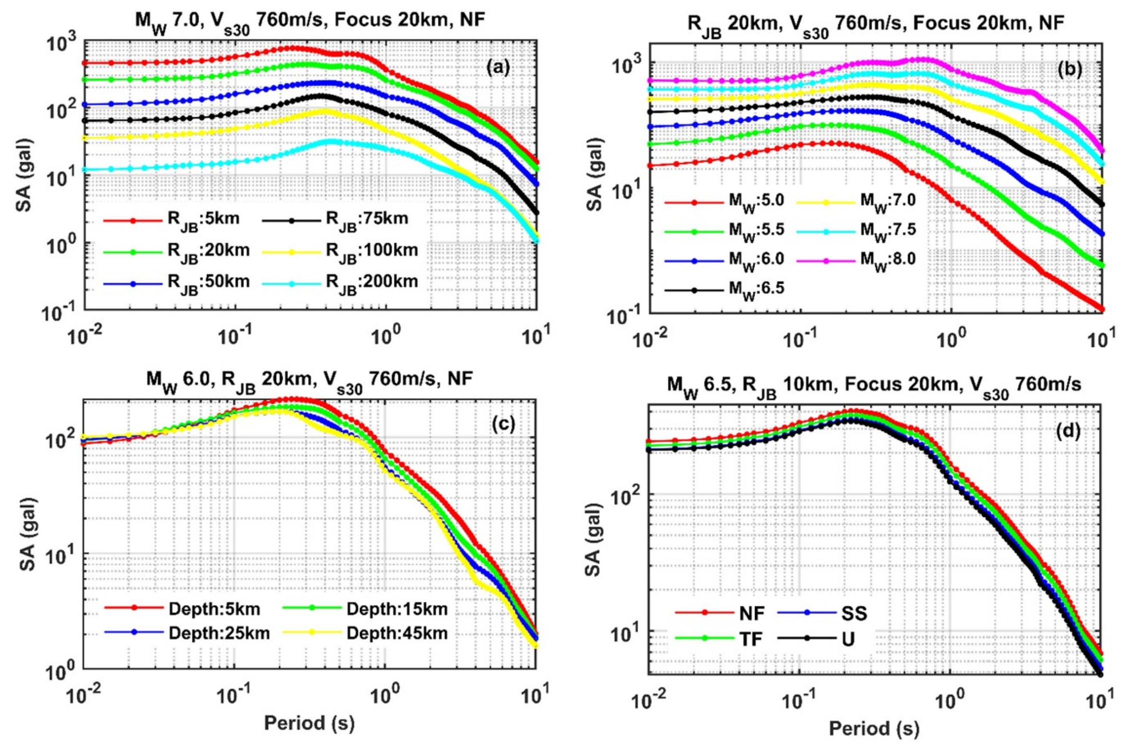


Figure 14. Variation of the response spectra with the seismological parameters.

saturation of the ordinate with magnitude is also evident. The distinct impact of the focal depth and fault mechanism is also evident in the lower panel of figure 14 (figures 14(c) and (d)).

Figure 15 presents the variation of the response spectra with V_{s30} across different magnitudes while keeping R_{JB} as 20 km, focal depth as 20 km, and normal faulting. A consistent pattern of soil amplification with all the bins indicates the model's capability to capture soil amplification effects. The model is further evaluated across four time periods: 1.5 s, 2.0 s, 5 s, and 10 s as illustrated in figure 16. For each period, variations in the median spectral ordinate with distance are plotted for different magnitudes, displayed in separate panels. These plots show the expected decay in spectral ordinates with distance and their scaling with magnitude across all periods. A similar comparison is presented in figure 17, illustrating how spectral ordinates vary with magnitude at different R_{JB}, leading to similar findings. Overall, these comparisons confirm the model's reliability in estimating mean spectral ordinates for horizontal components across all evaluated periods. The relative contribution of all the predictors to the PCs (specifically, their mean and sigma) in CVAE + ANN network and to the overall network are presented in figures 18(a) and (b), respectively. Distinct contribution of all the predictors is noted including the fault mechanism is observed which is not captured in case of NLPCA + ANN network.

7.1. Analysis of residuals and the standard deviation

The random effect residuals are calculated through mixed-effect regression, enabling an accurate decomposition of variability into inter-event, inter-site, and intra-event components. Below is a step-by-step procedure for developing the mixed-effects model adopted.

i. The model (12) is initially trained assuming no random effects.

$$y_{ij} = f(X_i, \beta) + \Delta_{ij},\ \Delta_{ij} \sim \mathbb{N}\left(0, \sigma^2\right) \tag{12}$$

y_{ij} denotes the observed MMI for the j^{th} recording of i^{th} event; X_i are the fixed-effect predictors; β is the vector of fixed-effect coefficients and Δ_{ij} is the residual error with σ as the total standard deviation. $\mathbb{N}(.)$ denoted normally distributed.

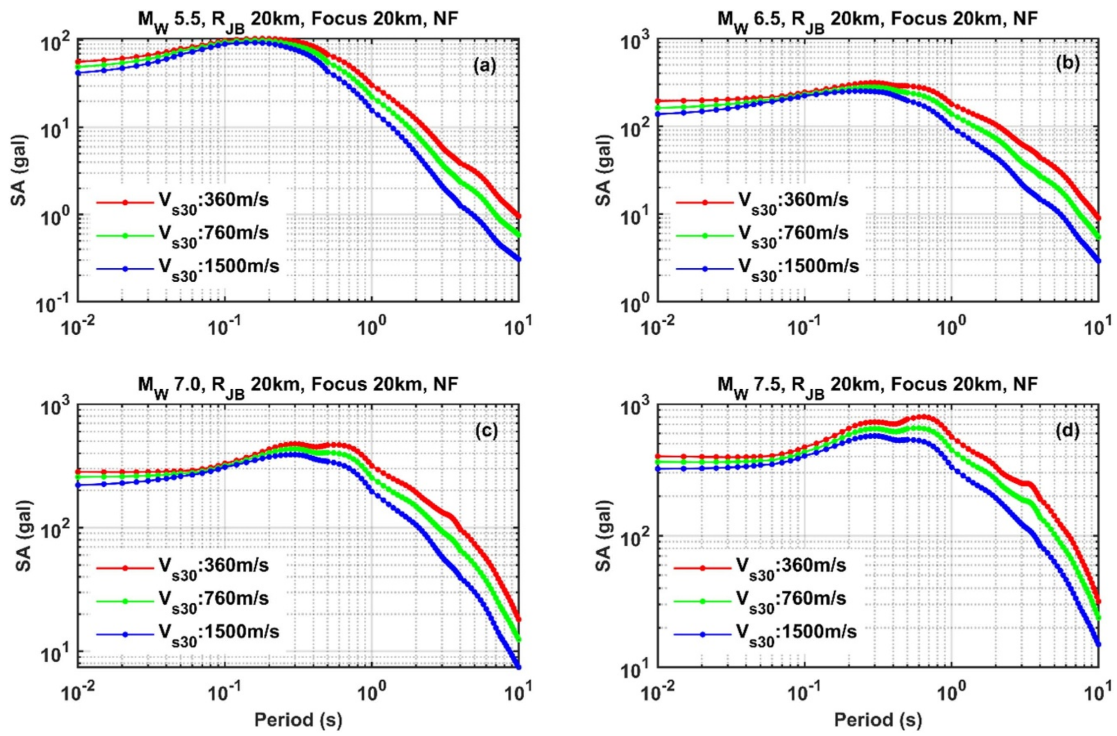


Figure 15. Variation of the response spectra with the average shear wave velocity (V_{s30}).

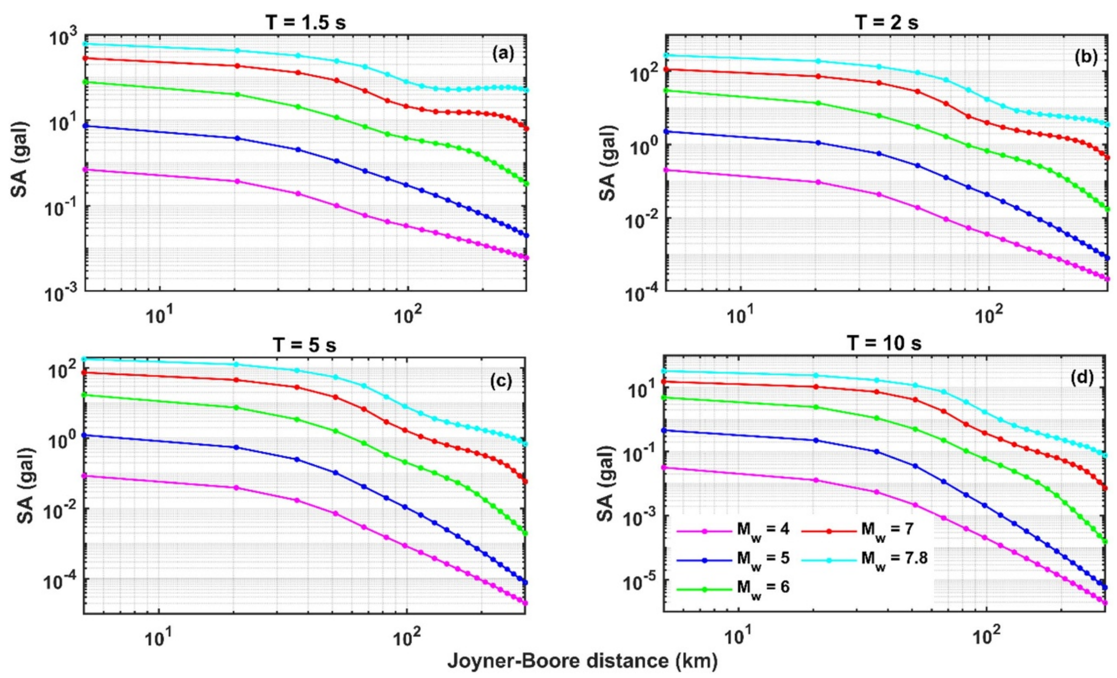


Figure 16. Variation of the spectral ordinate at different periods with the Joyner–Boore distance (R_{JB}).

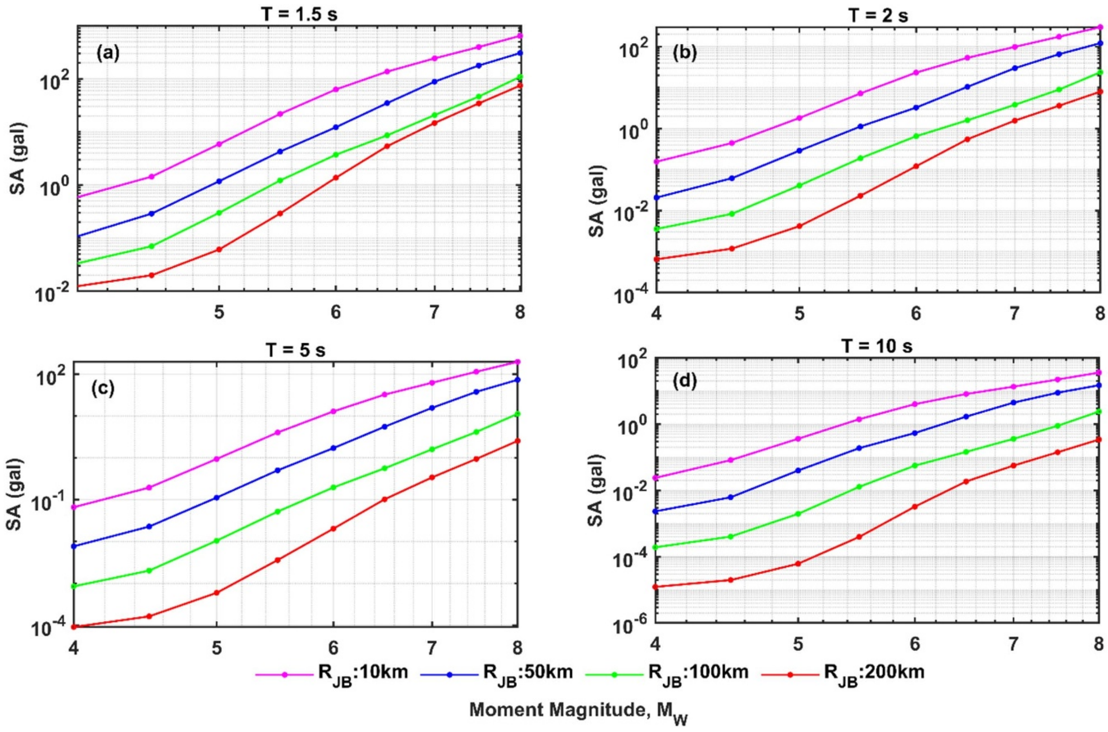


Figure 17. Variation of the spectral ordinate at different periods with the Moment magnitude (M_w).

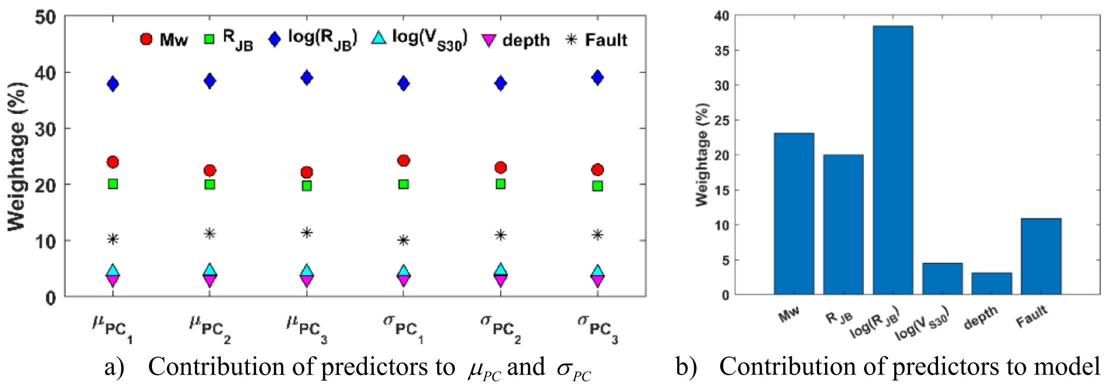


Figure 18. The relative impact of the input parameters in the CVAE+ ANN model.

ii. Residual components are computed by maximizing the log-likelihood function [logL (θ)] as per Abrahamson and Youngs (1992) and adjusted by Mohammadi *et al* (2023) as per equation (13).

$$\log\mathrm{L}(\theta) = -\frac{N}{2}\log(2\pi) - \frac{N-M}{2}\log\phi^2 - \frac{1}{2}\sum_{i=1}^{M}\log\left(\phi^2 + n_i\tau^2\right)$$
$$-\frac{1}{2\sigma^2}\sum_{i=1}^{M}\sum_{j=1}^{n_i}\left(y_{ij}-\mu_{ij}\right)^2 + \frac{\tau^2}{2\phi^2}\sum_{i=1}^{M}\frac{n_i^2}{\phi^2+n_i\tau^2}\left(\overline{Y_i}-\overline{\mu_i}\right)^2 \tag{13}$$
$$\overline{Y_i} = \frac{1}{n_i}\sum_{j=1}^{n_i} y_{ij};\ \overline{\mu_i} = \frac{1}{n_i}\sum_{j=1}^{n_i}\mu_{ij}$$

where, N is the number of records, M is the number of events, and n_i is the number of recordings for the i^{th} event. τ^2 is the variance of random effects for events, and ϕ^2 is the variance of the residual error.

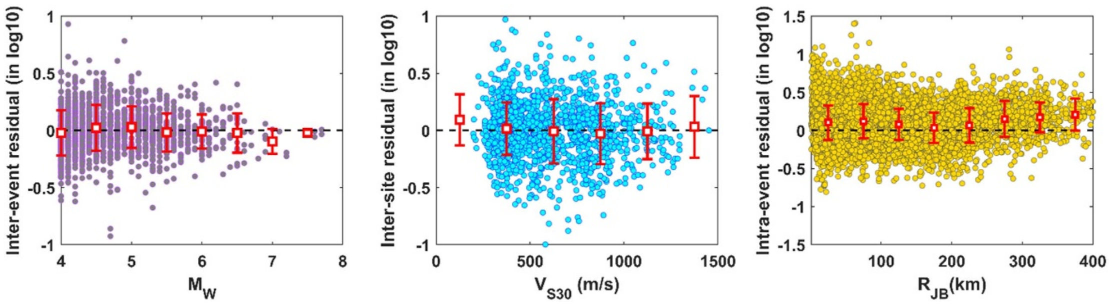


Figure 19. Variation of the residuals with seismological parameters for PGA [SA($T = 0$)].

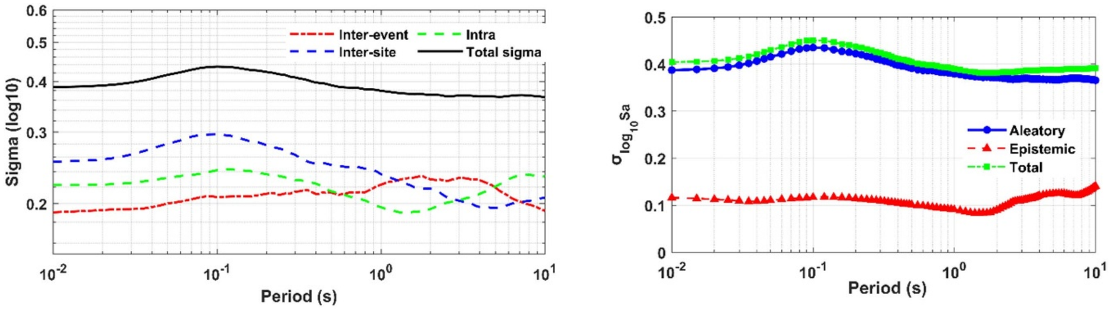


Figure 20. The standard deviation of the finalized CVAE model.

iii. Based on the estimated values of τ and ϕ along with the model parameters θ, the random-effect term is calculated using:

$$\eta_i = \frac{\tau^2 \sum_{j=1}^{n_i} \left(y_{ij} - \mu_{ij}\right)^2}{\phi^2 + n_i \tau^2} . \tag{14}$$

iv. iv. A new model is then trained using a fixed-effect procedure on the adjusted MMI $\left(y_{ij} - \eta_i\right)$.

v. Steps ii–iv are repeated until convergence, which terminates when the log-likelihood ratio, monitored via equation (15) falls below 0.5%, indicating convergence.

$$\mathrm{LHR} = \frac{\mathrm{LH}\,(ii) - \mathrm{LH}\,(ii-1)}{\mathrm{LH}\,(ii-1)} 100\%. \tag{15}$$

This process iteratively updates random effects and re-estimates fixed effects until convergence, capturing both inter-and intra-event variabilities.

The variation in these residuals with respect to predictor variables is illustrated in figure 19 only for $T = 0$ (PGA) for brevity. Each figure includes error bars representing the mean and one sigma bound for the residuals. It should be noted that all residuals are shown on a log10 scale. Regarding range, inter-event and inter-site residuals generally remain within ± 1, though a few intra-event residuals slightly exceed this bound for certain records. Importantly, these residual plots are observed to follow the Gaussian distribution and reveal no substantial bias relative to the predictor variables. The mean residuals remain nearly zero across all bins, suggesting the model is robust and well-calibrated. The sigma of the residuals across different bins is nearly similar, indicating a homoscedastic pattern. This consistent spread supports the model's stability across varying predictor values. To completely specify the distribution characteristics of the spectral ordinates, the standard deviation for each period associated with inter-event, inter-site, and intra as well as the total standard deviations are presented in the left panel of figure 20.

7.2. Epistemic uncertainty using CVAE

Revisiting the figure 13, it presents 100 ensemble response spectra predictions for sample ground motions, generated with the proposed CVAE model. The ensemble predictions provide insights into the epistemic

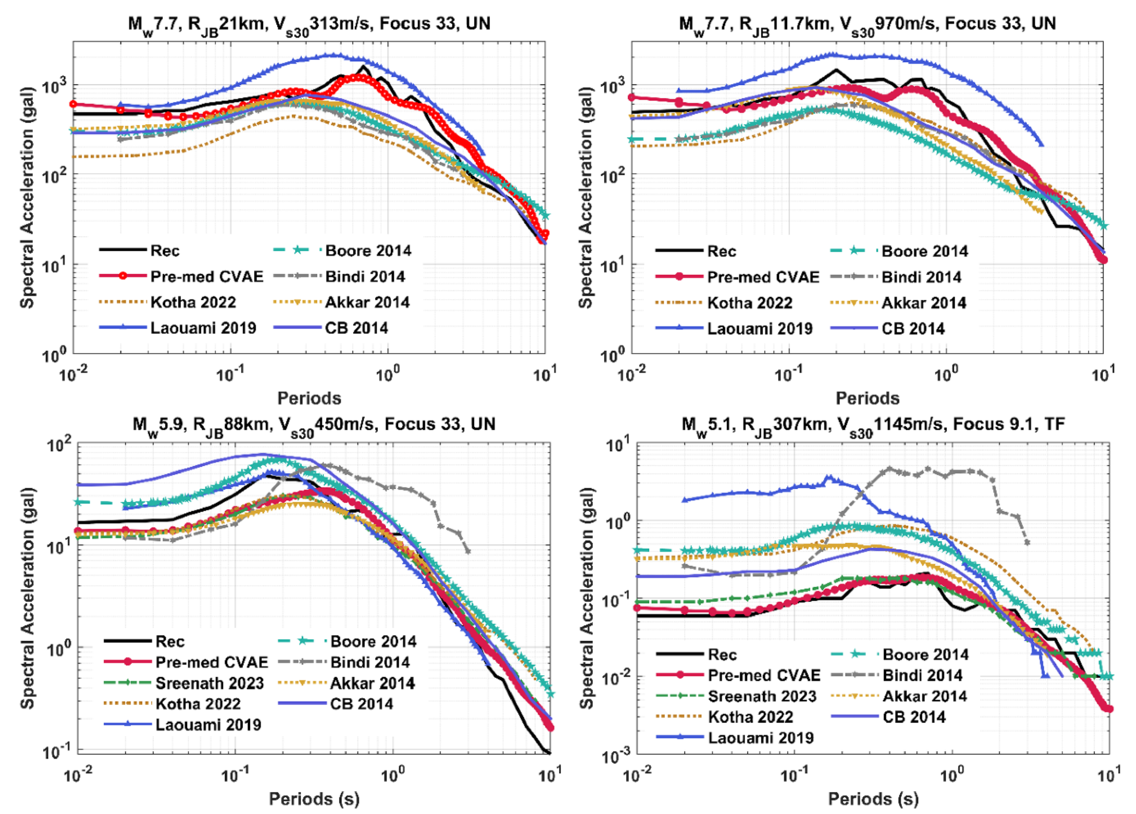


Figure 21. Comparison of the predicted spectra with the past models developed for the European region.

uncertainty associated with ground motions—a level of detail unattainable with point-estimate methods like shallow neural networks. Considering n^{th} ground motion, an ensemble of 'm' samples are generated. The standard deviation of this ensemble prediction in the epistemic standard deviation associated with n^{th} ground motion, denoted by $\sigma^2_{n-\text{epistemic}}$. The model's overall epistemic standard deviation, $\sigma_{\text{epistemic}}$, is then derived by averaging the epistemic variance of each ground motion as follows.

$$\sigma^2_{\text{epistemic}} = \frac{1}{T}\sum_{s=1}^{n}\sigma^2_{n-\text{epistemic}}. \tag{16}$$

Here, 100 predictions are generated per ground motion (i.e. $m = 100$), with a total of $n = 11\,675$ ground motions in the dataset. The mean epistemic standard deviation for individual ground motions ranges from 0.08 to 0.13 (in log10 scale), which is notably lower than the 0.35–0.48 range seen in ANN in the same scale. The total uncertainty, which includes both epistemic and aleatory components, is calculated as per equation (17) and is displayed in the right panel of figure 20, where both epistemic and aleatory uncertainties are included.

$$\sigma_{\text{total}} = \sqrt{\sigma^2_{\text{epistemic}} + \sigma^2_{\text{aleatory}}}. \tag{17}$$

8. Comparison with other GMMs

This section compares the proposed CVAE model with several established GMMs from recent literature, focusing on those developed for horizontal spectral ordinates in the European region, including models by Bindi *et al* (2014), Akkar *et al* (2014a), Laouami *et al* (2018), Kotha *et al* (2020, 2022), and Sreenath *et al* (2023). These models primarily utilize European and neighboring ground motion data. Specific models such as Bindi *et al* (2014) leveraged the RESORCE database for horizontal components, while Akkar *et al* (2014a) used pan-European data to develop a GMM for horizontal components. Laouami *et al* (2018) included data from Algeria, Europe, and the USA, enhancing regional accuracy. Kotha *et al* (2020, 2022) incorporated additional datasets and random effect adjustments, improving standard deviation estimates. Sreenath *et al* (2023) applied machine learning to a hybrid model with a limited magnitude range (3.2–7.4). Models such as Campbell and Bozorgnia (2014) and Bozorgnia and Campbell (2016) provide robust GMMs for horizontal spectral ordinates using PEER NGA-West2 data. Boore *et al* (2014) introduced a GMM for horizontal components. These models are included for comparison with the proposed BNN model due to their broad magnitude range and relevance to the European region.

Figure 21 compares the CVAE model's median predictions (denoted as Pre-med CVAE as it is the median of 100 ensemble predictions) for horizontal spectral acceleration with the prediction of the other models, capturing scenarios across varying depths, focal mechanisms, and distance ranges. Notably, the CVAE model aligns closely with recorded data (denoted as Rec), showing strong performance across near and far-field scenarios. While Bindi *et al* (2014) deviate significantly for lower magnitudes, Laouami *et al* (2018) tend to overestimate in most cases. The predictions of Kotha *et al* (2022) slightly underestimate the spectral ordinates at higher magnitudes. The CVAE model's results are close to Sreenath *et al* (2023), likely due to shared data with that used in this paper and as well due to the similarity in the model parameters. The results also show that the CVAE model aligns closely with Campbell and Bozorgnia (2014) (denoted by CB2014) and Boore *et al* (2014) across most records, suggesting potential global applicability.

9. Summary and conclusion

The study highlighted the algorithmic stability limitations of the NLPCA network and established CVAE as a more stable and robust alternative for GMM. Conventionally used for ensemble data reconstruction, CVAE was integrated with an ANN to develop a GMM for horizontal spectral ordinates. The model incorporated key seismological parameters, including earthquake magnitude, distance, shear wave velocity, focal mechanism, and focal depth, utilizing a large dataset from the Database (ESM 2.0).

The proposed CVAE + ANN framework effectively captured critical seismic characteristics such as magnitude scaling, ground motion attenuation, and nonlinear soil amplification. It demonstrated performance comparable to, and in some cases exceeding, existing state-of-the-art GMMs. The model exhibited strong predictive accuracy, as reflected by low RMSE (0.136) and MAE (0.183), along with a high R^2 (0.95), indicating reliable representation of data variability. Additionally, the MAPE (27%) was significantly lower than that of the NLPCA + ANN model (38%), underscoring CVAE's superior accuracy and enhanced spectral modeling reliability.

Beyond predictive accuracy, the CVAE-based approach improved the representation of ground motion variability by accurately modeling the standard deviations of spectral ordinates using mixed-effect regression. It also enabled the generation of ensemble outputs for given predictor variables, providing insights into epistemic uncertainty. The mean epistemic standard deviation for individual ground motions ranged from 0.08 to 0.13, which was notably lower than the 0.35–0.48 range of aleatory standard deviation in the CVAE network. These findings demonstrated CVAE's effectiveness in capturing complex seismic phenomena, thereby improving seismic hazard assessment accuracy.

It is important to note that the reported epistemic uncertainty is specific to the studied region and the range of seismological parameters considered. Further analysis may be necessary to establish a more generalized recommendation for epistemic uncertainty in GMMs.

Data availability statement

All data that support the findings of this study are included within the article (and any supplementary files).

Data and resources

ESM2.0 flatfile is available at https://esm-db.eu/#/products/flat_file and the database is available at https://esm.mi.ingv.it//flatfle-2018. Matlab code for generating ensemble acceleration response spectra predictions using the CVAE model is provided in https://figshare.com/s/b765d465efe16fbcd292.

Conflict of interest

The authors declare that there are no conflicts of interest regarding the publication of this paper.

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